

Objective:

Understand the New Keynesian framework through a simplified version from **Benigno (2015)**

Roadmap:

1. Households, Intertemporal Consumption and Aggregate Demand
2. Firms, Pricing and Aggregate Supply
3. Short Run and Long Run
4. AS-AD
5. PSET 7
6. Interpreting Policy Through the New Keynesian Model

The Model in One Slide

Agent	Problem	Delivers
Households	$\max u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$ $\text{s.t. } PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$	Euler equation → AD Labour supply → slope of AS
Firms	Monopolistic competition, $Y(j) = AL(j)$ A fraction α stuck at P^e in the short run	Pricing → AS
Central Bank	Sets the nominal rate i in the short run Anchors $\bar{p}$ in the long run	Shifts AD Nominal anchor $\bar{p}$
Government	$g, \bar{g}$, taxes $\{\tau_y, \tau_w, \tau_c, \bar{\tau}\}$, lump-sum T balancing its budget	Shifts AS + AD curves

Two periods:

- Short-run with **sticky prices**
- Long-run where **classical dichotomy** holds

The Household Problem

The household block from our earlier TA class, now with a nominal budget

$$\max_{C, L, \bar{C}, \bar{L}} u(C) - v(L) + \beta \{u(\bar{C}) - v(\bar{L})\} \quad (1)$$

$$\text{s.t.} \quad PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T \quad (2)$$

- Bars denote the *long run*. Two periods are enough for an intertemporal decision: the first is “today”, the second stands in for everything after prices have adjusted.
- The budget is *nominal*: a unit of consumption tomorrow costs $\bar{P}/(1+i)$ in today's money; a unit today costs P . The relative price of today versus tomorrow is $(1+i)P/\bar{P}$ — a *real interest rate*.
- T is lump-sum transfers, including firms' profits.

What is new?

Nothing in the household's problem.

What changes is what surrounds it: Some prices will not adjust to clear markets in the first period.

Two Margins, Two Curves

Within each period:

$$\frac{v_l(L)}{u_c(C)} = \frac{W}{P}$$

Consumption–Leisure tradeoff equals the real wage.

Tells firms labor costs at each level of activity:

Defines the **slope of AS**

Across periods: (Euler)

$$\frac{u_c(C)}{\beta u_c(\bar{C})} = (1+i) \frac{P}{\bar{P}} = 1+r \quad (3)$$

MRS between today's and tomorrow's consumption equals the real interest rate.

Tells us how much households want to spend today, given the future → Defines **AD**

Reading guide for the rest of the session

Every shift in AD is an Euler-equation story.

Every statement about the slope of AS is, in part, a labour-supply story.

From the Euler Equation to AD

1. Isoelastic utility, $u_c(C) = \frac{C^{1-\tilde{\sigma}}}{1-\tilde{\sigma}}$. Take logs of (3), and define $\rho \equiv -\ln \beta$:

$$\bar{c} - c = \tilde{\sigma}(r - \rho)$$

Consumption grows when the real rate exceeds impatience.

2. Real rate between periods is nominal rate minus inflation, $r = i - (\bar{p} - p)$, so:

$$\bar{c} - c = \tilde{\sigma} [i - \rho - (\bar{p} - p)]$$

3. Goods markets clear in both periods, $Y = C + G$ and $\bar{Y} = \bar{C} + \bar{G}$.

Log-linearised: $y = s_c c + g$, $\bar{y} = s_c \bar{c} + \bar{g}$.

Aggregate demand

$$y = \bar{y} + (g - \bar{g}) - \sigma[i - \rho - (\bar{p} - p)]$$

- Given the long run, higher prices today lead to lower output (negative AD).
- The mechanism is intertemporal substitution of consumption: $r = i - (\bar{p} - p)$.

Why AD Slopes Down

An intertemporal-substitution story, not a money-demand story

$$p \uparrow \Rightarrow \underbrace{\bar{p} - p}_{\text{expected inflation}} \downarrow \Rightarrow r = i - (\bar{p} - p) \uparrow \Rightarrow \text{saving } \uparrow, c \downarrow \Rightarrow y \downarrow$$

- **The anchor does the work.** A higher price level today, with tomorrow's price level given, means lower expected inflation — and so a higher real rate at the same nominal rate.
- **Slope $-1/\sigma$:** in (y, p) space. A large σ makes AD flatter: small price movements shift demand a lot.
- **No LM curve, no money stock, no investment:** The textbook AD slopes down through real balances and the interest rate that clears the money market; here it slopes down through the real rate households face (Benigno, 2015, sec. 5).

The instrument is inside the curve

i enters AD directly. The central bank does not move money and wait for i to follow: it sets i , and the real rate carries the effect to spending.

Monopolistic Competition

$$Y(j) = AL(j), \quad \Pi(j) = P(j)Y(j) - WL(j),$$

$$Y(j) = \left(\frac{P(j)}{P} \right)^{-\theta} (C + G)$$

- Firms are **no longer price-takers** — Operate under monopolistic competition, choosing prices given demand for their product/variety.
- **Price-setting friction:** A fraction α of firms is **unable to set prices**, having them at a given level P^e and adjusting production accordingly.
- A firm free to choose sets a **mark-up over marginal cost**

$$P(j) = \frac{\theta}{\theta - 1} \frac{W}{A}$$

Benigno's (10)–(11) also carry sales and payroll taxes; they return in the fiscal section as part of the labour wedge.

Marginal Cost Rises With Output

Where the household's labour supply enters the firm's pricing Replace W/P with the household's intratemporal condition:

$$\frac{P(j)}{P} = \frac{1 + \mu}{A} \frac{v_l(L)}{u_c(C)} = \frac{1 + \mu}{A} \frac{L^\eta}{C^{-1/\tilde{\sigma}}} \quad (12)$$

- **Real marginal cost = real wage / A = the household's MRS / A.** More output needs more hours ($L \uparrow$, through η) and brings more consumption ($C \uparrow$, through $1/\tilde{\sigma}$); both raise the wage households ask for.
- So a firm free to reprice wants a higher *relative price* when the economy is booming. This is the whole source of the upward slope of AS.
- μ is the aggregate **mark-up**: the wedge between price and marginal cost, $1 + \mu = \frac{\theta}{\theta - 1}$ without taxes. A rise in μ is a **mark-up shock**.

Sticky Prices and the AS Curve

1. α firms keep P^e , chosen on information prior to the shocks (sticky prices, or equally sticky information). The other $1 - \alpha$ choose $\tilde{P}$. The price index averages them:

$$p = \alpha p^e + (1 - \alpha) \tilde{p}$$

2. For the adjusters, (12) in logs, measured against natural output y_n — what output would be if *all* firms could adjust:

$$\tilde{p} - p = (\tilde{\sigma}^{-1} + \eta)(y - y_n)$$

3. Eliminate $\tilde{p}$:

Aggregate supply

$$p - p^e = \kappa(y - y_n), \quad \kappa \equiv \frac{(1 - \alpha)(\tilde{\sigma}^{-1} + \eta)}{\alpha}$$

When output is above its natural level, marginal cost is high, the firms that can reprice raise their prices, and the price level ends up above what was expected.

Reading AS

- **Position.** AS always passes through (p^e, y_n) . Prices deviate from what firms expected only when output deviates from its natural level.
- **Slope κ has two pieces.** $(\tilde{\sigma}^{-1} + \eta)$ is how fast real marginal cost rises with output — a property of preferences, from labour supply. $(1 - \alpha)/\alpha$ is how many firms can act on it — a property of price setting.
- **Limits.** $\alpha \rightarrow 0$: $\kappa \rightarrow \infty$, AS vertical at y_n , prices absorb everything. $\alpha \rightarrow 1$: $\kappa \rightarrow 0$, AS flat at p^e , output absorbs everything.
- **What moves it:** anything that moves y_n (productivity, public spending, mark-ups — next section) or p^e (Fig. 2).

Not yet a New Keynesian Phillips curve

Unexpected prices and no $E_t \pi_{t+1}$: this is the New Classical Phillips curve. Phelps (1967) and Lucas (1973) derive the same equation from imperfect information rather than sticky prices. An equation does not identify its own microfoundation.

Two Periods, Two Regimes

	Short run	Long run
Prices	α fixed at P^e , $1 - \alpha$ flexible	All flexible
AS	Upward sloping, slope κ	Vertical at $\bar{y}_n$
Output	Can differ from y_n	Equal to $\bar{y}_n$
Monetary policy	Sets i — not neutral	Sets $\bar{p}$ — neutral
Fiscal policy	Real effects (g , taxes)	Real effects ($\bar{g}$, taxes)

The classical dichotomy, in the second period only

In the long run, real variables are set by preferences, technology, mark-ups and fiscal policy; the central bank chooses only the price level $\bar{p}$. Every bit of monetary non-neutrality in the model lives in $\alpha > 0$, in the first period.

Natural Output: The Flexible-Price Benchmark

$$y_n = \frac{1 + \eta}{\tilde{\sigma}^{-1} + \eta} a + \frac{\tilde{\sigma}^{-1}}{\tilde{\sigma}^{-1} + \eta} g - \frac{1}{\tilde{\sigma}^{-1} + \eta} \mu \quad (15)$$

- **Productivity $a \uparrow$:** each hour produces more, $y_n \uparrow$.
- **Public spending $g \uparrow$:** less private consumption raises the marginal utility of income, so households work more. y_n rises, but by less than g — consumption is crowded out. One-for-one only if $\eta = 0$.
- **Mark-up $\mu \uparrow$** (monopoly power or any tax): the wedge between price and marginal cost works like a tax on work, $y_n \downarrow$.
- The same formula with bars gives $\bar{y}_n$, and $\bar{c}_n = s_C^{-1}(\bar{y}_n - \bar{g})$ is the long-run consumption that sits inside today's AD.

Natural is not the same as desirable

y_n is the flexible-price equilibrium, distortions included. Whether it is a good place to be is a separate question.

Natural Versus Efficient Output

Two benchmarks that answer two different questions

Natural

$$y_n = \frac{1 + \eta}{\tilde{\sigma}^{-1} + \eta} a + \frac{\tilde{\sigma}^{-1}}{\tilde{\sigma}^{-1} + \eta} g - \frac{\mu}{\tilde{\sigma}^{-1} + \eta}$$

What would a flexible-price *market economy* produce?

Efficient

$$y_e = \frac{1 + \eta}{\tilde{\sigma}^{-1} + \eta} a + \frac{\tilde{\sigma}^{-1}}{\tilde{\sigma}^{-1} + \eta} g$$

What maximises $u(C) - v(L)$ subject to $Y = AL$ and $Y = C + G$?

$$y_n - y_e = -\frac{\mu}{\tilde{\sigma}^{-1} + \eta}$$

Keep for later

The welfare-relevant target is y_e ; the AS curve is anchored at y_n . For every shock we will ask which of the two it moves — that is what decides whether the central bank faces a choice.

The Natural Rate of Interest

Where the short run meets the long run

The real rate $i - (\bar{p} - p)$ at which AD delivers exactly $y = y_n$:

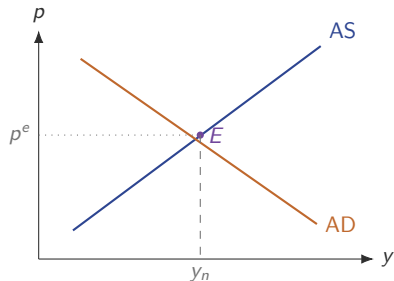
$$r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) + \sigma^{-1}(g - \bar{g}) + (\bar{\tau}_c - \tau_c)$$

- r_n is the rate at which households want to buy the natural level of output. It comes from AD, with $y = y_n$.
- r_n is high when the future looks better than the present ($\bar{y}_n > y_n$: smoothing pulls spending forward, so a high rate is needed to hold it back) or when public spending is temporarily high.
- r_n depends on preferences, technology, mark-ups and fiscal policy — not on the central bank. The central bank sets i ; it can only choose where the actual real rate sits relative to r_n .

The starting point of every exercise

$\bar{p} = p^e$ and $i = r_n$: prices are where firms expected them, and AD crosses AS at y_n .

The Initial Equilibrium



$$\text{AS: } p - p^e = \kappa(y - y_n)$$

- Slope κ , through (p^e, y_n) by construction.

$$\text{AD: } y = \bar{y} + (g - \bar{g}) - \sigma[i - \rho - (\bar{p} - p)]$$

- Slope $-1/\sigma$. With $\bar{p} = p^e$ and $i = r_n$ it crosses y_n at p^e too.

At E:

Zero output gap, prices as expected. Every exercise is a displacement from E .

Solving the Model: One Gap Drives Everything

Substitute AS into AD and use the definition of r_n Let $\hat{r} \equiv i - (\bar{p} - p^e)$: the real rate policy sets, evaluated at the prices firms expected. Then

Reduced form

$$y - y_n = -\frac{\sigma}{1 + \sigma\kappa}(\hat{r} - r_n), \quad p - p^e = \kappa(y - y_n)$$

- **Only the real-rate gap $\hat{r} - r_n$ matters.** Shocks move r_n (and y_n); policy moves $\hat{r}$ — through i today and $\bar{p}$ tomorrow.
- Setting $\hat{r} = r_n$ gives $y = y_n$ and $p = p^e$ at once. Closing the gap and stabilising prices are the same task — as long as y_n is the right target.
- **The split.** A demand impulse D at given prices moves output by $D/(1 + \sigma\kappa)$; the rest is choked off by higher prices raising the real rate. The larger κ (fewer sticky firms), the more goes into p .

Reading a Shock in Four Steps

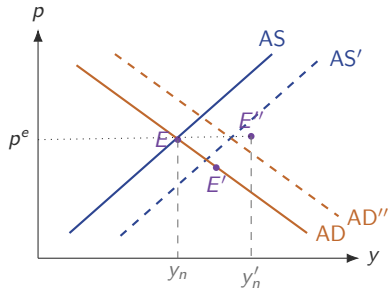
1. **Which equation does it enter?** AS, through y_n or p^e ; AD, through i , g , τ_c , $\bar{p}$, or the long run $(\bar{y}_n, \bar{g}, \bar{\tau})$.
2. **Which benchmarks move?** y_n , $\bar{y}_n$, y_e — and therefore r_n .
3. **The new intersection with i unchanged.** Signs of $y - y_n$, $p - p^e$, and $y - y_e$.
4. **What policy can do.** Is there an i that puts the economy at y_e with $p = p^e$? If not, what does each option cost?

Discipline

Draw the shift before telling the story, and tell the story through one household decision (save or spend, work or not) and one firm decision (reprice or not).

Worked Example: A Temporary Productivity Gain

$a \uparrow$ today, $\bar{a}$ unchanged (Fig. 6)



- **AS shifts right:** $y_n \uparrow$, and y_e by the same amount. Marginal cost falls; flexible firms cut prices.
- **AD does not move:** the long run is unchanged. But r_n falls — today is temporarily good relative to the future.
- $E \rightarrow E'$: $p \downarrow$ raises expected inflation $\bar{p} - p$, the real rate falls, consumption rises — but not by enough: y rises by less than y_n , a negative gap.
- **Policy:** cut i until AD passes through (p^e, y'_n) . At E'' the gap is closed and prices are stable.